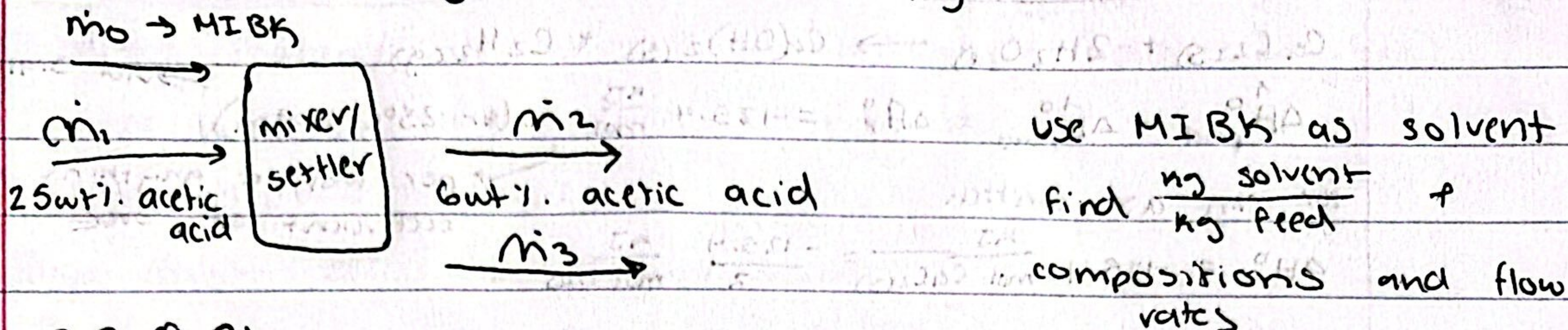


HW #9 ~~Oil and Water~~ ~~margin~~ ~~8~~

1.) 5.40 - fig 5.15 is on pg 421



~~PRO Feed~~

for product, it needs to be water rich, so it is left most point on graph.

at 6 wt% acetic acid, left most point gives

~ 2.5 wt% MIBK. so, $100 - (6 + 2.5) = 91.5$ wt% water

$$\dot{m}_1 = \dot{m}_2 + \dot{m}_3 \quad \text{AA bal: } .25 \dot{m}_1 = .06 \dot{m}_2 + \dot{m}_3 X_{3,AA}$$

$$.25(\dot{m}_2 + \dot{m}_3) = .06 \dot{m}_2 + X_{3,AA} \dot{m}_3$$

$$.19 \dot{m}_2 = X_{3,AA} \dot{m}_3$$

for $\dot{m}_3$, look at right most point $X_{3,AA}$

~ 92 wt% MIBK, ~ 4.5 wt% acetic acid, so

$$100 - (92 + 4.5) = 3.5 \text{ wt\% water}$$

water balance

$$.75 \dot{m}_1 = .915 \dot{m}_2 + .035 \dot{m}_3$$

using basis of $\dot{m}_1 = 100 \text{ kg/s}$

$$.75(100) = .915 \dot{m}_2 + .035 \dot{m}_3$$

$$.25(100) = .06 \dot{m}_2 + .045 \dot{m}_3$$

$$75 = .915 \dot{m}_2 + .035(555.6 - 1.33 \dot{m}_2)$$

$$25 - .06 \dot{m}_2 = .045 \dot{m}_3$$

$$55.56 = .868 \dot{m}_2$$

$$\dot{m}_3 = 555.6 - 1.33 \dot{m}_2$$

$$\dot{m}_2 = 63.98 \text{ kg/s}$$

$$\dot{m}_3 = 555.6 - 1.33(63.98)$$

$$\dot{m}_3 = 470.29 \text{ kg/s}$$

$$.025 \dot{m}_2 + .92 \dot{m}_3 = \dot{m}_0$$

$$.025(63.98) + .92(470.29) = 434.26 \text{ kg/s} = \dot{m}_0$$

$$\frac{\text{kg solvent}}{\text{kg feed}} = \frac{\dot{m}_0}{\dot{m}_1} = \frac{434.26}{100} = 4.34 = \frac{\text{kg solvent}}{\text{kg feed}}$$

$$\dot{m}_2 = 63.98 \text{ kg/s}$$

$$\dot{m}_3 = 470.29 \text{ kg/s}$$

$$X_{2,AA} = .06$$

$$X_{3,AA} = .045$$

$$X_{2,W} = .915$$

$$X_{3,W} = .035$$

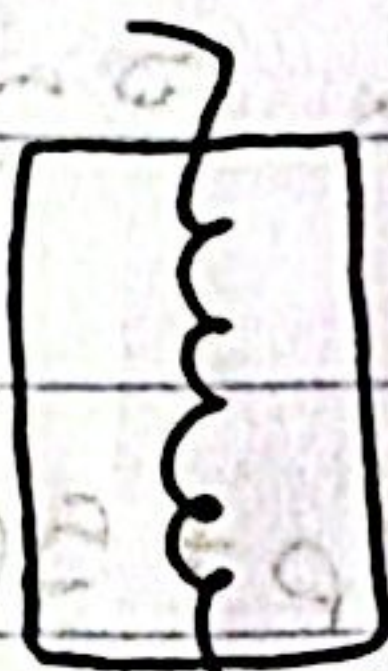
$$X_{2,M} = .025$$

$$X_{3,M} = .92$$

3.) - im skipping 2 for now

air
25°C → 150°C

~~sh = 3640 J/mol~~



$$\dot{v}_c = 1.25 \frac{\text{m}^3}{\text{min}}$$

$$P = 122 \text{ kPa}$$

$$T = 150^\circ\text{C}$$

$$\Delta h = 3640 \text{ J/mol}$$

$$a.) \Delta H + \Delta \cancel{P_E} + \Delta \cancel{P_F} = \dot{Q} + \dot{W}_S$$

$$\cancel{dm^3} = L$$

$$\Delta H = \dot{Q}$$

at inlet, ~~T = 150°C~~ T = 25°C

$$PV = nRT$$

at outlet, P = 122 kPa, T = 150°C

$$P\dot{V} = nRT$$

$$\left(\frac{122 \text{ kPa}}{101 \text{ kPa}} \right) \left(\frac{1.25 \text{ m}^3}{\text{min}} \cdot \frac{1 \text{ L}}{.001 \text{ mL}} \right) = n \left(.082057 \frac{\text{atm} \cdot \text{L}}{\text{mol} \cdot \text{K}} \right) (423 \text{ K})$$

$$\dot{n}_2 = 43.5 \frac{\text{mol}}{\text{min}} \times \frac{1 \text{ min}}{60 \text{ s}} = .725 \text{ mol/s}$$

$$\dot{Q} = n \cdot \Delta H = .725 \frac{\text{mol}}{\text{s}} \cdot 3640 \frac{\text{J}}{\text{mol}} = 2639.02 \text{ J/s} = 2.639 \text{ kW}$$

b.) you can't tell, not enough information. You would need - at least velocity

2.) C_p is mmol/L q_p is mmol/g

One liter of wastewater, containing 1g Ph w/ 5g C.

$y = 0.2426x + .7599$ - from Freundlich

MM: 94.11

$$\frac{1g}{94.11g} = .0106 \text{ mol} = \underline{1.06 \text{ mmol}}$$

$$\frac{106 \text{ mmol}}{1 \text{ L}}$$

$$\ln(q_p) = .2426(\ln(106)) + .7599$$

$$q_p = 6.6280 \quad q_p = 3.793$$

$$\frac{3.793}{10.6} = .358$$

$$3.793 \cdot 5 =$$

$$\ln(q_p) = .2426(\ln(C_p)) + .7599$$

$$C_p = .726 \frac{\text{mmol}}{\text{L}} \quad q_p = 1.978 \frac{\text{mmol}}{\text{g}} \times 5g = 9.89$$

$$C_p + 5q_p = 10.6$$

$$C_p + 5(e^{C_p \cdot .2426 + 2.13})$$

$$\ln(q_p) = .2426 \ln(C_p) + .7599$$

$$q_p = e^{C_p \cdot .2426 + 2.13}$$

$$\ln(q_p) = .2426 \ln(10.6 - 5q_p) + .7599$$

$$\ln(7.379p) = 13.5 + 2.1$$

$$e^{\frac{10.6 - 2.1 \cdot 5}{10.6}} = .991$$

$$q_p = 2.1$$

4) 1gmol of H_2O

$$\Delta H + \Delta E_k + \Delta E_p = \dot{Q} + \dot{W}_s$$

$$\frac{1}{2} m v^2 = \dot{Q} = \frac{1}{2} (1 \text{ gmol}) (100 \frac{\text{km}}{\text{h}})^2 = \dot{Q}$$

$$\frac{1}{2} \left(\frac{1 \text{ gmol}}{18.02} \times 18.02 \right) \left(\frac{100 \text{ km}}{\text{h}} \times \frac{1 \text{ hour}}{3600 \text{ s}} \right)^2 = \dot{Q}$$

$$\frac{1}{2} (18.02) \left(\frac{100 \text{ km}}{\text{h}} \times \frac{1 \text{ h}}{60 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ s}} \times \frac{1000 \text{ m}}{1 \text{ km}} \right)^2 = \dot{Q}$$

$$\dot{Q} = 6.952.16 \text{ J}$$

table
B.16

part c,
B.3

~~part c~~

$$b.) \Delta E_p = \dot{Q} = (1 \text{ g mol} \cdot \frac{18.02 \text{ g}}{\text{mol}}) (9.8) (100)$$

$$\boxed{\dot{Q} = 176596 \text{ J}}$$

$$c.) \Delta u = \Delta h + p \Delta v$$

$$\Delta u = \Delta h + \delta p \Delta v$$

$$\Delta u = (2776 - 2783) + (1.697 - 1.714) \left(\frac{100 \text{ kPa}}{101 \text{ kPa}} \cdot 1 \text{ atm} \right) V_1 = 1.697$$

$$\boxed{\Delta u = -7.617 \text{ J}}$$

$$pV = nRT \quad V_1 = \left(\frac{1}{18.02} \right) (0.082) (373)$$

$$\left(\frac{10 \text{ kPa}}{101 \text{ kPa}} \cdot 1 \text{ atm} \right) V_2 = \left(\frac{1}{18.02} \right) (0.082) (373)$$

$$d.) 0 \rightarrow 100^\circ \text{C} \text{ at } 5 \text{ bar}$$

$$V_2 = 17.143$$

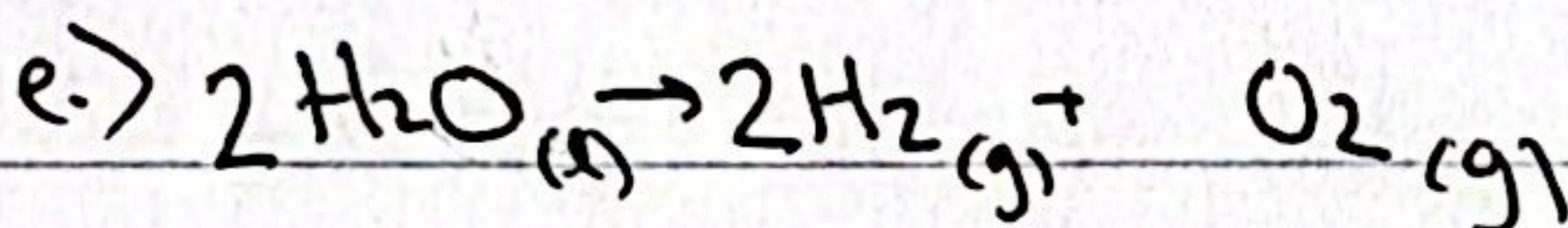
$$\Delta u = \Delta h + p \Delta v$$

$$\Delta u = (419.51 - 100) + 5 (1.69 - 1.24)$$

$$\boxed{\Delta u = 319.96}$$

$$pV = nRT \quad p = 5$$

$$V_2 = \frac{\left(\frac{1}{18.02} \right) (0.082) (373)}{5}$$



$$V_1 =$$

at 25°C

$$\Delta H^\circ_{\text{H}_2} = 0$$

$$\Delta H^\circ_{\text{O}_2} = 0$$

$$\Delta H^\circ_{\text{H}_2\text{O}} = -285.84 \text{ kJ/mol}$$

$$\Delta H = -285.84 \text{ kJ} = -285840 \text{ J} \times 2 \text{ coz } 2 \text{H}_2$$

$$\Delta H^\circ_r = \Delta H^\circ_p - \Delta H^\circ_r = -2 \cdot (-285840)$$

$$\boxed{\dot{Q} = 571680 \text{ J}}$$

$$f.) E = mc^2$$

$$E = \left(\frac{1}{18.02} \right) (3 \times 10^8)^2$$

$$\boxed{E = 4.99 \times 10^{15} \text{ J}}$$

19/25